

No. of Functional Features Included in the Solution

Date	02 July 2026
Team ID	
Project Name	Rising Waters – A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Functional Features Overview

This document lists all functional features included in the Rising Waters solution, each described with its purpose, implementation status, and contribution to the overall solution.

S. No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Historical Rainfall EDA	National rainfall distribution, seasonality, trend and regional comparison charts (1901-2015)	Notebook	Done	15%
2	Data Pre-processing Pipeline	Missing-value imputation, IQR outlier capping, feature scaling	Notebook	Done	10%
3	Multi-Model Training & Comparison	Decision Tree, Random Forest, KNN, XGBoost trained and evaluated with confusion matrices & classification reports	Notebook	Done	20%
4	Automatic Best-Model Selection	Best classifier chosen by test accuracy and persisted with scaler + feature names	Notebook	Done	10%
5	Home Page with Live Stats	Displays active model name and accuracy, feature count, and how-it-works walkthrough	App	Done	10%
6	Guided Prediction Form	10 grouped, labelled input fields (Atmospheric / Seasonal Rainfall / Derived Indicators) with units, help text and defaults	App	Done	15%
7	Input Validation	Per-field numeric validation with inline error messages on invalid input	App	Done	5%
8	Flood-Risk Result Pages	Dedicated Flood Chance and No Flood Chance pages with probability and next-step guidance	App	Done	10%
9	Multi-Target Deployment	Docker, IBM Cloud Foundry manifest, and generic Procfile deployment options	Deployment	Done	5%

Feature Summary

Metric	Count / Value
Total Features Planned	9

Total Features Implemented	9
Core / Must-Have Features	6 (EDA, Pre-processing, Model Comparison, Best-Model Selection, Prediction Form, Result Pages)
Nice-to-Have Features	3 (Live home-page stats, input validation UX, multi-target deployment)